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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 16: Theorems Related with Area

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - all four theorems of the unit (16.1 to 16.4) with their proofs in simple steps, the useful corollaries, fully worked solutions of the book exercise, and exam practice with answers.

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> **The one idea behind this whole unit:** area depends only on the **base** and the **altitude**. If two figures share the same base (or equal bases) and have the same altitude, their areas are equal - even when their shapes look completely different. Learn to spot "same base, same altitude" in a figure and every theorem and proof in this unit becomes easy.

What this unit covers

1. Area, base and altitude - the key ideas
2. Theorem 16.1 - parallelograms on the same base, between the same parallels
3. Theorem 16.2 - parallelograms on equal bases, same altitude
4. Theorem 16.3 - triangles on the same base, same altitude
5. Theorem 16.4 - triangles on equal bases, same altitude, and corollaries
6. How to use the theorems in proofs - Exercise 16.1 Question 1 worked fully
7. Worked examples - midpoint fractions, a parallel-line proof, and tiling
8. Theorem summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 16 (book pages 291-297). The notes follow the book's theorems, exercise and review exercise. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Area, Base and Altitude - The Key Ideas

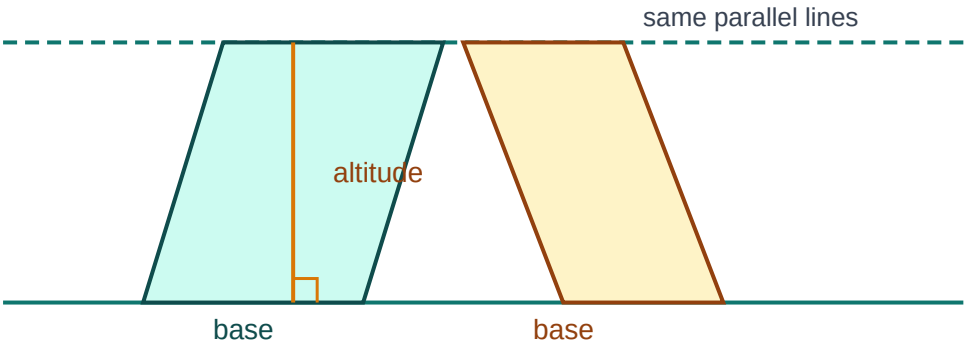
Area measures the surface inside a figure. It is always written in square units, such as cm^2, m^2 or square feet. Two figures can have exactly the same area and still have different shapes, different sides and different angles.

Base and altitude

- Any side of a triangle or parallelogram can be chosen as its **base**.
- The **altitude** is the perpendicular (at 90 degrees) from the opposite vertex - or from the opposite side - down to the line containing the base. Altitude is a perpendicular distance, never a slanting side.
- Area of a parallelogram = **base x altitude**.
- Area of a triangle = **1/2 x base x altitude**.

Between the same parallel lines

The perpendicular distance between two parallel lines is the **same everywhere** - it never changes. So every figure standing on one of the lines with its top on the other line has exactly the same altitude. In the book's words, figures "lying between the same parallel lines" are figures "of the same altitude". Whenever a question says *same altitude*, look for the two parallel lines in the figure.

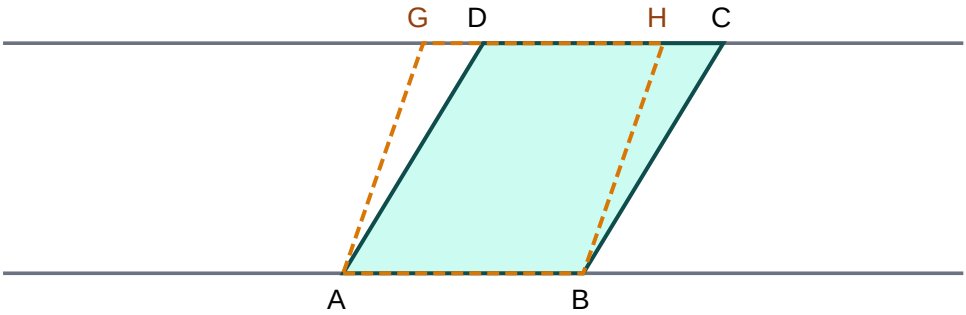


Both figures stand between the same two parallel lines, so both have the same altitude - the perpendicular distance between the lines.

> **Equal in area does not mean congruent.** Congruent figures are identical in shape and size. Figures of equal area only cover the same amount of surface. A long thin parallelogram and a wide short one can have equal areas without being congruent. The book states this clearly on page 291, and it is a favourite MCQ.

2. Theorem 16.1 - Parallelograms on the Same Base

Theorem 16.1: Parallelograms on the same base and lying between the same parallel lines (or of the same altitude) are equal in area.



In the book's figure the parallelograms are ABCD and ABGH on the common base AB, between the parallel lines XY and PQ.

Given: Parallelograms ABCD and ABGH stand on the same base AB and lie between the same parallel lines, so they have the same altitude.

To prove: Area of parallelogram ABCD = Area of parallelogram ABGH.

Proof in easy steps

Statement	Reason
In triangles ADH and BCG: AH = BG	Opposite sides of a parallelogram
Angle at D in triangle ADH = angle at C in triangle BCG, and angle at H = angle at G	Corresponding angles (the top vertices lie on the same parallel line)
Therefore triangle ADH is congruent to triangle BCG	A.A.S. congruence
So the two triangles are equal in area	Congruent figures have equal areas
Subtract each triangle in turn from the same whole figure ABCH	Equals taken from equals
The remainders are parallelograms ABCD and ABGH, so they are equal in area	Proved

Numeric check: base 10 cm, altitude 6 cm. Both parallelograms have area 10 x 6 = **60 cm^2**, no matter how much each one slants. Slanting changes the shape, never the area.

3. Theorem 16.2 - Parallelograms on Equal Bases

Theorem 16.2: Parallelograms on equal bases and having the same altitude are equal in area.

The bases no longer need to be the *same* base - they only need to be **equal in length**. The two parallelograms may stand in completely different places on the same pair of parallel lines.

Given (book figure): Parallelograms ABCD and EFGH lie between the same parallel lines (same altitude) and AB = EF.

To prove: Area of ABCD = Area of EFGH.

Proof idea - build a bridge

1. **Construction:** join A to H and B to G, making a third parallelogram ABGH.
2. HG = EF (opposite sides of parallelogram EFGH) and EF = AB (given), so AB = HG. Also AB is parallel to HG, since both lie on the two parallel lines. Hence ABGH really is a parallelogram.
3. ABCD and ABGH share the base AB and the altitude, so they are equal in area by **Theorem 16.1**.
4. ABGH and EFGH share the base HG = EF and the altitude, so they are also equal in area by **Theorem 16.1**.
5. Therefore Area of ABCD = Area of EFGH (transitive property - both equal the area of ABGH).

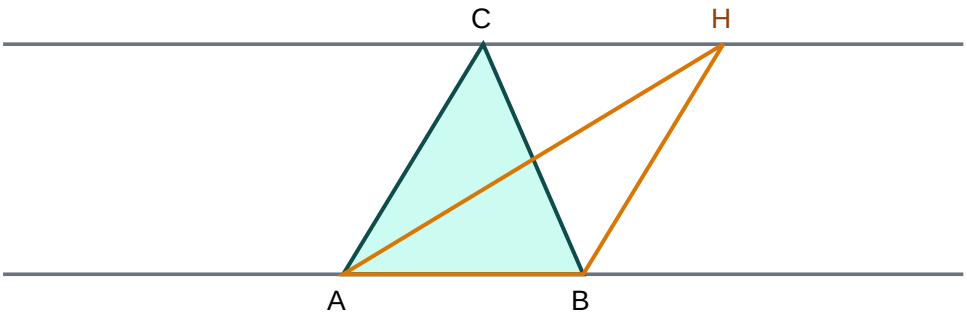
Example - checking with numbers. Parallelogram P has base 8 cm and altitude 5 cm. Parallelogram Q, in a different place between the same parallels, also has base 8 cm and the same altitude 5 cm.

Area of P = 8 x 5 = 40 cm^2. Area of Q = 8 x 5 = 40 cm^2. Equal, exactly as Theorem 16.2 says.

Both conditions are needed. Equal bases alone are not enough - if the altitudes differ, the areas differ. Equal bases *and* the same altitude together give equal areas.

4. Theorem 16.3 - Triangles on the Same Base

Theorem 16.3: Triangles on the same base and of the same altitude are equal in area.



Triangles ABC and ABH share the base AB. Their third vertices C and H both lie on the same parallel line, so the altitude is the same.

Given (book figure): Triangles ABC and ABH have the same base AB and the same altitude, lying between the parallel lines PQ and XY.

To prove: Area of triangle ABC = Area of triangle ABH.

Proof in easy steps

1. **Construction:** complete each triangle to a parallelogram - draw AD parallel to BC and BG parallel to AH, meeting the top line at D and G. Now ABCD and ABGH are parallelograms.
2. The two parallelograms stand on the same base AB with the same altitude, so they are equal in area by **Theorem 16.1**.
3. A diagonal divides a parallelogram into two equal triangles, so Area of triangle ABC = 1/2 x Area of parallelogram ABCD, and Area of triangle ABH = 1/2 x Area of parallelogram ABGH.
4. Halves of equal areas are equal. Therefore the two triangles are equal in area.

> **Picture it:** slide the top vertex C anywhere along the top parallel line. The triangle stretches and leans, but base and altitude never change, so its area never changes.

5. Theorem 16.4 and Useful Corollaries

Theorem 16.4: Triangles on equal bases and of the same altitude are equal in area.

Given (book figure): Triangles ABD and KLM lie between the same parallel lines PQ and XY, and their bases are equal: AB = KL.

To prove: Area of triangle ABD = Area of triangle KLM.

Proof in easy steps

1. **Construction:** complete the parallelograms - draw BC parallel to AD and KN parallel to LM, meeting the top line at C and N. ABCD and KLMN are parallelograms.
2. Their bases AB and KL are equal and their altitude is the same, so the parallelograms are equal in area by **Theorem 16.2**.
3. Each triangle is half of its parallelogram (a diagonal bisects a parallelogram), so Area of triangle ABD = 1/2 x Area of ABCD and Area of triangle KLM = 1/2 x Area of KLMN.
4. Halves of equal areas are equal. Therefore the triangles are equal in area.

Corollaries - the results you will use most

Result	Why it is true
A diagonal of a parallelogram divides it into two triangles of equal area; each is half the parallelogram.	The two triangles are congruent, and together they make the whole.
A triangle and a parallelogram on the same base and between the same parallels: the triangle is **half** the parallelogram.	Triangle = 1/2 x base x altitude; parallelogram = base x altitude.
A **median** of a triangle divides it into two triangles of equal area.	The median creates two triangles on equal bases (the two halves of the base) with the same altitude - Theorem 16.4. This is Review Exercise Question 2.
Triangles with the same altitude have areas in the ratio of their bases; triangles on the same base have areas in the ratio of their altitudes.	Area = 1/2 x base x altitude, so only the changing part matters.

6. Using the Theorems in Proofs

Almost every proof in this unit follows the same four steps:

1. Find two triangles or parallelograms that share a **base**, or have equal bases.
2. Show they have the **same altitude** - usually because their vertices lie on a line parallel to the base.
3. Quote the matching theorem (16.1, 16.2, 16.3 or 16.4) to get equal areas.
4. If needed, **add or subtract the same figure** from both equal areas to reach the pair the question asks about.

Worked proof - Exercise 16.1, Question 1 (from the book)

In triangle ABC, PQ is drawn parallel to BC, cutting AB and AC at P and Q. BQ and CP meet at R. Prove the pairs in (i) to (iv) are equal in area.

(i) Triangles PBC and QBC. Both stand on the same base BC, and their vertices P and Q lie on the line PQ, which is parallel to BC. Same base, same altitude - equal in area by **Theorem 16.3**.

(ii) Triangles BRP and CRQ. From (i), subtract the common triangle BRC from both: Area PBC - Area BRC = Area QBC - Area BRC. The remainders are exactly triangles BRP and CRQ, so they are equal in area.

(iii) Triangles PQB and PQC. Both stand on the same base PQ, and their vertices B and C lie on BC, which is parallel to PQ. Same base, same altitude - equal in area by **Theorem 16.3**.

(iv) Triangles ABQ and ACP. From (iii), add the common triangle APQ to both: Area PQB + Area APQ = Area PQC + Area APQ. The totals are triangles ABQ and ACP, so they are equal in area.

> **Exam tip:** parts (ii) and (iv) show the two standard moves - *subtract a common part* and *add a common part*. Write the theorem name and number in every proof; board marking gives steps marks for it.

7. Worked Examples from Exercise 16.1

Example 1 - Midpoint fractions (Exercise 16.1, Q2). PQRS is a parallelogram. A and B are the mid-points of PQ and PS. What fraction of the area of the parallelogram is each triangle?

Take the area of PQRS as 1 whole (check with base 4 and altitude 4, area 16, if you like numbers).

- **Triangle PAB = 1/8.** Its base PA is half of PQ and its altitude from B is half the altitude of the parallelogram (B is a mid-point), so area = 1/2 x (1/2 base) x (1/2 altitude) = 1/8 of base x altitude.
- **Triangle QBR = 1/2.** Check it with coordinates: place P at (0,0), Q at (1,0), S at (0,1) and R at (1,1), so the parallelogram has area 1. Then B is (0, 1/2), and the area of triangle QBR works out to exactly 1/2.
- **Triangle BAR = 3/8.** In the same coordinate check, A is (1/2, 0), B is (0, 1/2) and R is (1, 1), and the area of triangle BAR works out to 3/8.

Answers: (i) 1/8 (ii) 1/2 (iii) 3/8 of the area of the parallelogram.

Example 2 - Proving lines parallel (Exercise 16.1, Q3). ABCD is a quadrilateral; diagonals AC and BD meet at E. If triangles ABE and CDE are equal in area, prove AD is parallel to BC.

Add triangle AED to both equal areas: Area ABE + Area AED = Area CDE + Area AED, so Area ABD = Area ACD. These two triangles stand on the same base AD and have equal areas, so they have the same altitude - meaning B and C are at the same perpendicular distance from AD, on a line parallel to AD. Hence AD is parallel to BC.

Example 3 - Tiling a floor (Exercise 16.1, Q4). How many tiles, each 8 inches square, pave a rectangular space 18 x 30 feet?

Area of floor = 18 x 30 = 540 square feet = 540 x 144 = 77,760 square inches. Area of one tile = 8 x 8 = 64 square inches. Number of tiles = 77,760 / 64 = **1,215 tiles**.

8. Summary and Common Mistakes

The four theorems - learn these word for word

- **16.1** Parallelograms on the same base and lying between the same parallel lines (same altitude) are equal in area.
- **16.2** Parallelograms on equal bases and having the same altitude are equal in area.
- **16.3** Triangles on the same base and of the same altitude are equal in area.
- **16.4** Triangles on equal bases and of the same altitude are equal in area.

- Altitude = perpendicular distance to the chosen base. The distance between parallel lines is constant.
- Area of parallelogram = base x altitude. Area of triangle = 1/2 x base x altitude.
- A diagonal halves a parallelogram. A median halves a triangle. A triangle on the same base and altitude as a parallelogram is half of it.
- Equal area does **not** mean congruent.

Common mistakes

Mistake	Correction
Saying two figures are equal in area from the same base alone	The altitude must also be the same - check the parallel lines.
Writing "equal in area, therefore congruent"	Equal area says nothing about shape or sides. Never write this.
Using a slanting side as the altitude	Altitude is always perpendicular to the base.
Forgetting the 1/2 for a triangle	Triangle area = 1/2 x base x altitude, every time.
Subtracting different parts from two equal areas	Only adding or subtracting the *same* (common) figure keeps areas equal.
Mixing units in area sums (feet with inches)	Convert first - 1 square foot = 144 square inches, as in the tiling example.

9. Practice - MCQs and Questions

These MCQs are the book's Review Exercise 16. Choose one answer for each.

1. The perpendicular distance between two lines is the same everywhere. The lines are: (a) perpendicular (b) parallel (c) intersecting (d) none of these
2. If two triangles have equal area, they will _____ be congruent as well: (a) not necessarily (b) necessarily (c) definitely (d) none of these
3. The perpendicular from a vertex of a triangle to its opposite side is called: (a) median (b) perpendicular bisector (c) altitude (d) angle bisector
4. Parallelograms having the same base and the same altitude are _____: (a) congruent (b) equal in area (c) similar (d) all of these
5. Two parallelograms have equal bases. They will have the same area if: (a) their altitudes are equal (b) their altitude is the same (c) they lie between the same parallel lines (d) all of these
6. If two triangles have equal bases and equal altitudes, what else will they have equal? (a) area (b) perimeter (c) size (d) angles
7. A triangle has base 4 feet and height 4 feet. Its area is: (a) 4 sq ft (b) 8 sq ft (c) 16 sq ft (d) impossible to find
8. A square has a diagonal of 10 units. The area of the square is: (a) 25 (b) 50 (c) 100 (d) impossible to find
9. Find the area of a triangle with base 15/4 inches and altitude 8/5 inches: (a) 2 sq in (b) 6 sq in (c) 3 sq in (d) 4.35 sq in

Short and proof questions

1. Prove that a median of a triangle divides it into two parts of equal area.
2. In triangle ABC, D and E are the mid-points of AB and AC. Prove that triangles ADE, BDE and CDE are equal in area.
3. P is any point on the median AD of triangle ABC. Prove that triangles ABP and ACP are equal in area.
4. Find the area of a triangle with base 20 inches and altitude 12 inches.
5. A parallelogram has base 12 cm and altitude 7 cm. A second parallelogram between the same parallels has base 12 cm. Find its area, and the area of a triangle on the same base and altitude.

> **Answer key - MCQs:** 1 (b) 2 (a) 3 (c) 4 (b) 5 (d) 6 (a) 7 (b) 8 sq ft 8 (b) 50 9 (c) 3 sq in.
>
> **Answer key - questions:** Q1-Q3 are proofs: use Theorem 16.4 (equal bases from the median / mid-points, same altitude), then subtract equal parts where needed. Q4: 1/2 x 20 x 12 = **120 square inches**. Q5: parallelogram = 12 x 7 = **84 cm^2** (Theorem 16.2); triangle = half of it = **42 cm^2**.